

Arjuna JEE (2025)

Mathematics

Straight Lines

DPP: 1

- Q1** The points $A(0, -1)$, $B(2, 1)$, $C(0, 3)$, $D(-2, 1)$ are the vertices of a:
 (A) square (B) rectangle
 (C) parallelogram (D) none of these
- Q2** The point $(5, -1)$ divides the line segment joining points A and B in the ratio $2 : 3$. If A is $(11, -3)$, then B will be-
 (A) $(4, -2)$
 (B) $(4, 2)$
 (C) $(-4, 2)$
 (D) $(-4, -2)$
- Q3** If the line $2x + y = k$ passes through the point which divides the line segment joining the points $(1, 1)$ and $(2, 4)$ in the ratio $3 : 2$, then k is equal to
 (A) $\frac{29}{5}$
 (B) 5
 (C) 6
 (D) $\frac{11}{15}$
- Q4** If $P(1, 2)$, $Q(4, 6)$, $R(5, 7)$ and $S(a, b)$ are the vertices of a parallelogram $PQRS$, then
 (A) $a = 2, b = 4$
 (B) $a = 4, b = 4$
 (C) $a = 2, b = 3$
 (D) $a = 3, b = 5$
- Q5** The points with co-ordinates $(2a, 3a)$, $(3b, 2b)$ and (c, c) are collinear -
 (A) For no value of a, b, c
 (B) For all values of a, b, c
 (C) If $a, c/5, b$ are in H.P
 (D) If $a, 2c/5, b$ are in H.P.
- Q6** The line segment joining the points $(-3, -4)$ and $(1, 2)$ is divided by y -axis in the ratio
 (A) $2 : 3$
 (B) $1 : 3$
 (C) $3 : 1$
 (D) $3 : 2$
- Q7** Values of k for which the points $(k, 2 - 2k)$, $(1 - k, 2k)$ and $(-4 - k, 6 - 2k)$ are collinear, will be
 (A) $1, -1/2$
 (B) $1, 1/2$
 (C) $-1, 1/2$
 (D) $-1, -1/2$
- Q8** The triangle whose vertices are $(8, 2)$, $(5, -3)$, $(0, 0)$ will be
 (A) isosceles
 (B) equilateral
 (C) right angled
 (D) Both (A) and (C)
- Q9** $A(6, 3)$; $B(-3, 5)$; $C(4, -2)$ and $D(x, 3x)$ are four points. If the areas of $\triangle DBC$ and $\triangle ABC$ are in the ratio $1 : 2$, then x is equal to
 (A) $11/8$ (B) 3
 (C) $8/11$ (D) None of these
- Q10** If P, Q, R are collinear points such that $P(7, 7)$, $Q(3, 4)$ and $PR = 10$, then R is
 (A) $(1, 1)$
 (B) $(1, -1)$
 (C) $(-1, 1)$
 (D) $(-1, -1)$
- Q11** $A(3, 4)$ and $B(5, -2)$ are two given points. If $PA = PB$ and area of $\triangle PAB = 10$, then P is-
 (A) $(7, 1)$ (B) $(7, 2)$
 (C) $(-7, 2)$ (D) $(-7, -1)$
- Q12**



ABC is an isosceles triangle. If the coordinates of the base are $B(1, 3)$ and $C(-2, 7)$, the coordinates of vertex A can be

- (A) $(1, 6)$
- (B) $(-1/2, 5)$
- (C) $(5/6, 6)$
- (D) none of these

Q13 If $(2, -3)$, $(6, 5)$ and $(-2, 1)$ are three consecutive vertices of a rhombus, then its area is

- (A) 24
- (B) 36
- (C) 18
- (D) 48

Q14 The area of the quadrilateral formed by points

$(a^2 + 2ab, b^2)$, $(a^2 + b^2, 2ab)$,

$(a^2, b^2 + 2ab)$

and $(a^2 + b^2 - 2ab, 4ab)$ is-

- (A) Zero
- (B) $(a + b)^2$
- (C) $a^2 + b^2$
- (D) $(a - b)^2$



Answer Key

Q1 (A)
Q2 (C)
Q3 (C)
Q4 (C)
Q5 (D)
Q6 (C)
Q7 (C)

Q8 (D)
Q9 (A)
Q10 (C)
Q11 (B)
Q12 (C)
Q13 (D)
Q14 (A)



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